The Vehicles, the Network and the Market: What Lunar ATC Puts Inside Each Vehicle and How Its Marketplace Works

Azizjon Rahimov

Roatify Research · Preprint, October 2026 · Code, data and pre-registration: github.com/azizjonrahimov/Lunar-Haul · Live site: azizjonrahimov.github.io/Lunar-Haul

Abstract. The rovers, hoppers and landers planned for a lunar south-pole base in 2028–2032 have different owners, carry far less than the base needs, and share no radio standard. Our companion papers found that a robot-run job market beats Earth dispatch. This paper asks what it needs in each vehicle, whether that fits their computers and radios, and how its money works among owners who distrust one another. We collected the planned vehicles' published capabilities with their provenance and described the radio network: LunaNet, a robot mesh on 5 m terrain, Earth hidden two weeks a month. We measured our implementation at one, four and eight times the fleet, scaled it to flight hardware, and priced four payment rules over 1,080 simulated years for individual rationality, bid shading and audits. Scaled by 54.8 to the reference flight processor, the safety check takes 41.8 % of one core at the worst class and the radio traffic is 7.68 kbit/s for 20 robots, within target; one auction round in a hundred takes 1.12 s, above the one-second decision. Every owner beat its alone profit under every rule, but owner C fell below the no-payment market under every paid rule; the bid rules ran a deficit, the posted price broke even. First-price bid shading gained at most 7.7 M USD a year with an interval including zero, so that clause of H5 is not supported; the second price reduced it (supported); the Shapley clause is supported. A post hoc split shows the first-price gain's cash part positive, +2.6 to +5.4 M USD, and its delay part noisy. Vehicle designers can budget, to within the stated factor, one vehicle's share of a lunar traffic and market network, and base operators can choose a payment rule priced in budget and in who pays whom.

Keywords: lunar surface operations, multi-owner robot fleets, mechanism design, surface radio network, runtime assurance, flight software sizing

1. Introduction

The vehicles that will drive, hop and land at a lunar south-pole base between 2028 and 2032 are known by name. NASA ordered two crew rovers in May 2026, Astrolab's CLV-1 and Lunar Outpost's Pegasus, and stated in October 2026 what they will do: carry up to 80 kg of samples and tools, drive 10 km on a charge and 6 km a day when driven from Earth [1]. Cargo-class rovers that could move a tonne or two exist only as company designs [2, 3]. Four hopping drones of about 250 kg arrive in 2028 [1, 7], a small cargo rover of 250 kg is a catalogue product [3], and the landers of the period put between 150 kg and 3 t on the surface at a time [11, 12, 13, 14]. NASA's own planning puts the base's recurring need at 2,500–10,000 kg a year and single elements at 500–15,000 kg, and says that demand and mobility capacity are mismatched by 1,000 to 15,000 kg per asset [5, 6]. The fleet of the period is small, its vehicles are owned by different companies, and no two of them are built to talk to each other.

The radio that would connect them is not standardised either. The LunaNet Interoperability Specification fixes the links between orbit and the surface and between the Moon and Earth, routes IP and bundle-protocol traffic from surface users, and marks surface Wi-Fi and cellular links as not part of LunaNet 1.0, with their physical layer “bilaterally defined” [15]. The one cellular network flown to the Moon ran for about 25 minutes on a lander

lying on its side and never completed its call [17]. NASA's surface-relay concept reaches about 10 km from a 10 m mast where terrain allows, and its viewshed study found a single 4 m tower covering 10.6–23.7 % of a 20 × 20 km polar area [16]. Earth itself is hidden from the pole for about 14 days of every month [16, 19]. Any coordination among these vehicles has to run on the vehicles, over short, terrain-blocked links, with Earth as an occasional witness rather than a controller.

Our two companion papers argued that this is enough. The first put a job market on the robots, run over a mesh computed from 5 m terrain with the base's real Earth windows, and found that it keeps the gain from sharing vehicles across owners through every blackout [60]. The second gave the robots rules of the road and a small safety program that overrides any planner, checked the rules exhaustively on a discrete model and in simulated encounters, and priced them in cargo delay [61]. Both papers took three things for granted: that each vehicle can carry the software and the radio the design needs, that a base computer and its spare can hold the shared reservation table and the ledger, and that owners will accept a market in which their robots bid against one another for work. This paper is about those three assumptions. We call the design Lunar ATC, by analogy with air-traffic control, though its decisions are made by the vehicles and a base computer on the Moon rather than by controllers.

Each assumption has a literature that stops short of our case. Multi-robot autonomy for flight exists in one system, CADRE, three rovers of one owner that elect a leader over a mesh and run a central planner on it, with a lightweight distributed database [26, 27, 28]; its radio band, rate and range are not public [10]. Field robotics in denied communication, the DARPA Subterranean Challenge, settled on mesh radios, droppable relays, data mules and time budgets out of contact, and the winning team fielded single-robot autonomy only [54, 55]. Runtime-assurance filters have measured computation costs, with implicit filters 11–13 times costlier than explicit switching on a docking problem [45], but no one has sized a safety program for a lunar vehicle's computer. Mechanism design says that efficiency, truthful bidding, individual rationality and a balanced budget cannot all be had at once [29, 30, 31], and freight carriers who collaborate share the gain by Shapley value and name a fair rule and a trusted coordinator as their main obstacles [35, 36]; ride-hailing data put the cost of fragmentation among operators at 2.5–67 % of fleet per extra operator, falling with demand density [39]. Multi-vendor robot fleets negotiate traffic in practice in Open-RMF, which reports high processor load during negotiation and has no stable wire protocol [41]. Robotics task allocation otherwise assumes cooperative robots; the closest work on self-interested robots is about paths, not jobs [34]. We found no study that sizes the on-vehicle software and radio of a multi-owner robot network against the vehicles that will actually fly, and none that measures a robot market's payments, fairness and manipulability in a logistics model.

Our question is therefore in three parts. What do the vehicles of 2028–2032 carry, and what can they not carry at all? What does Lunar ATC put inside each vehicle, how much processor time, memory, radio bandwidth and power does it need as implemented, and what happens when the base computer fails? And under which payment rules is every owner better off in the shared market than alone, how much can an owner gain by misreporting its costs, and what do audits from Earth recover?

We answered the first part from published sources, with every figure flagged by provenance and the contradictions between sources recorded. We answered the second by instrumenting our own implementation: one function decides every move and reservation, one auction runs the market, one message layer carries bids, grants, status and ledger entries over the mesh, and one base computer with a spare holds the shared table and the ledger. We measured these on the laptop that ran the companion experiments, at one, four and eight times today's fleet, and scaled the results to a radiation-tolerant processor by a stated factor; we swept the base computer's

failure rate and measured how long the fleet is without a leader at an assumed re-election delay. We answered the third with a market experiment, E46, in which three owners with equal and with 50/35/15 shares of the cargo run the pooled market under four payment rules over 30 seeds, and in which one owner scales its bids by 0.8 to 1.25 while the others bid honestly, with Earth charging the expected penalty of an audit of 0, 5 or 20 % of the jobs afterwards. The hypotheses, H5 on incentives and H10 on sizing, were written down before any run [59].

The results are these. Lunar ATC fits the vehicles of Section 3.2 as implemented: at the worst class the safety check takes 41.8 % of one core of the reference processor, the on-robot part 0.4 MB, and 20 robots generate 7.68 kbit/s against a 16 kbit/s-class link (Sections 5.1 to 5.3). In the market, every rule made every owner at least as well off as alone in both splits, though owner C was worse off than under no payments under every paid rule; the operator ran a deficit under first price, a deficit under the second price and a balanced budget under the posted price (Sections 5.5 to 5.7). H5's first clause, that first-price bids reward shading, is not supported; its second, that the second price reduces the gain, is supported; its third, that the posted price with Shapley sharing leaves no owner worse off than alone, is supported (Section 5.7).

Our contributions are:

- **A provenance-flagged account of the lunar vehicles of 2028–2032 and the surface radio network**, with the capacity plateau the fleet implies and the contradictions between sources recorded rather than resolved.
- **A bill of materials for Lunar ATC inside one vehicle**: the safety program, the planner, the radio unit and the local copies of the reservation table and ledger, with processor time, memory, bandwidth, power and failover measured on our implementation and scaled to flight hardware with a stated factor.
- **Market rules for owners who do not trust one another**, with the money flow from bid to settlement on Earth, and a measurement of individual rationality, manipulation by bid scaling and audits under four payment rules.

Several things are outside this paper. Which vehicle takes which job, and how the robots' auction behaves under message loss and Earth blackouts, is the subject of [60]; the rules of the road and the proof that the safety program keeps them are the subject of [61]; here both are treated as given and only their cost in computation, memory and bandwidth is measured. The vehicles' internal layouts are published for few of them, and this paper describes the parts Lunar ATC adds, not the vehicles' own avionics. Owners who misreport their position or availability, hog chargers or pick easy jobs are not simulated; the one manipulation tested is bid scaling. The market has three owners; Section 7 says what that leaves open.

2. Related work

2.1 The vehicles and what they carry

NASA's 2024 mobility white paper states the gap in plain numbers: the base's elements weigh 500–15,000 kg and move up to 5,000 m, “current demand and mobility capacity are mismatched on the order of 1,000 to 15,000 kg per asset”, wheel and soil interactions do not scale linearly with vehicle size, and the cooperation of several robots on one load “may be an enabling feature” and a reason to standardise robot interfaces [5]. The same series puts the recurring logistics need at 2,500–10,000 kg a year with occasional deliveries of 12,000–15,000 kg [6]. The original Lunar Terrain Vehicle requirement asked for at least 800 kg of uncrewed cargo [5]; what NASA bought in May 2026 was two crew-sortie rovers of about 907 kg (CLV-1) and a lighter, arm-less

Pegasus, and what it stated in October 2026 was 80 kg of samples and tools, 10 km per charge and 6 km a day when driven from Earth [1, 4]. The cargo-class parents of those rovers, FLEX with 2,000 kg on two top-deck interfaces and Eagle with a cargo bed and an arm, are company claims without a contract [2, 3]. Below them sit a 250 kg rover with a 200 kg payload [3], hoppers of about 250 kg that fly about 50 km in 14 days [7] and a 1 m hopper that carries 8–10 kg over 25 km [8]. The one two-rover cargo carry on record shares 475 kg between two 800 kg rovers with a formation error of 9.2 cm mean and 33.4 cm maximum, and does not describe the inter-rover link [25]. No battery size is published for any Lunar Terrain Vehicle; the reference points are VIPER's 5,420 Wh [9] and the Apollo rover's two 121 Ah silver-zinc batteries [64]. Our vehicle table (Section 3.2) is built from these sources; its lesson for this paper is that the fleet's capacity is a plateau set by two uncontracted cargo rovers, and that everything heavier moves only if two vehicles, usually of two owners, lift it together [60]. The policy frame for several owners on one ridge is thin. The Artemis Accords' safety zones ask for notification and coordination, not exclusion [52]; NASA's policy analysis of landing and surface operations finds transit corridors near the pole as narrow as 20–40 m and says that once three or more actors operate, some centralised coordination may become necessary [51]; and the one proposal for lunar surface traffic regulation raises priority of passage, passing and signalling without giving rules [53].

2.2 Surface radio

LunaNet standardises proximity links from orbit, direct-with-Earth links, the network layer (IPv4, IPv6 and Bundle Protocol version 7, which is mandatory for delay-tolerant traffic) and the time and navigation signal; it marks “IP over Wi-Fi” and “IP over 3GPP” on the surface as not part of LunaNet 1.0, lists the lunar cellular bands as under investigation, and leaves sub-IP details of surface links “out of scope … bilaterally defined” [15]. Its S-band proximity rates run from 0.5 ksps at the low end to 5 Msps [15]. The delay-tolerant networking that carries the Bundle Protocol became an operational service on NASA's near-space and deep-space networks in January 2026, with no demonstration on the lunar surface yet [22]. NASA's Lunar Surface Relay concept carries cellular sector antennas on a 10 m boom for coverage “out to 10 km, depending on local terrain”, Wi-Fi at about 300 m, two rubidium clocks for surface timing, and a stationary version whose communications draw 131 W; its viewshed study of the Malapert massif found a 4 m tower covering 10.6–23.7 % of a 20 × 20 km area and a 20 m tower 12.1–29.2 %, with only 7 of 400 candidate sites meeting both of its constraints, and warned that moving a tower by about 1 km “could have severe impacts” on coverage [16]. Lander-to-payload Wi-Fi at 2.4 GHz is flown on Griffin and Peregrine [18]. Nokia's cellular network on IM-2 targeted a few hundred metres to 2–5 km with antennas 3–5 m high and ran for about 25 minutes without completing a call, so no measured lunar cellular range or rate exists [17]. CADRE's three rovers carry mesh radios with ultra-wideband ranging, but their band, rate and range are not published [10]. Direct-to-Earth coverage at candidate south-pole sites is 17–43 % of a year, 80–86 % with one relay [19], and the pole does not see Earth for about 14 days a month [16]. The physics behind these numbers is geometry: on a smooth Moon of radius 1,737.4 km the horizon of an antenna at height h is $\sqrt{(2Rh)}$, 2.95 km at 2.5 m and 5.89 km at 10 m, with no atmospheric refraction [63]; rover antennas cannot clear the first Fresnel zone at kilometre ranges (5.6 m radius at 2.4 GHz over 1 km), so links sit near grazing, and a blocked path loses about 6 dB at grazing and 20 dB at a Fresnel parameter of 2.4 under the standard knife-edge model [20]; path loss over regolith “can be much more severe than the free space propagation” and the south-pole regolith's permittivity is unmeasured [21]. Lunar clocks run 56.02 µs a day fast relative to Earth, which a shared time base absorbs [23]. In short, the robot mesh of Lunar ATC has no flight precedent at its scale, and the design has to accept links that fail in steps as a vehicle drives behind a ridge.

2.3 Flight software on robots that coordinate

CADRE is the one multi-robot autonomy system prepared for lunar flight: three rovers and a base station on the lander, a centralised planner on an elected leader with distributed execution, a lightweight distributed database, and a mesh radio; the leader is chosen by a spanning-tree election that first finds a unique appointer and then the healthiest agent, and is re-elected on every 30-minute wake from a cooling shutdown; on a leader change, in-progress tasks stop and the new leader replans [26, 27, 28]. It is one owner and science exploration, not logistics, but its leader election and its database are the pattern our base-computer failover follows. The DARPA Subterranean Challenge shows how teams coped with denied communication: mesh radios with four to six deployed nodes, effective ranges of 49–1,058 m, maximum delays of 1–13 s, robots acting as relays and data mules, and a warning that requiring ordered delivery of mission-critical topics built large queues after reconnection [54]; the winning team ran out-of-contact exploration on a time budget and deployed single-robot autonomy only [55]. Open-RMF lets fleets of unrelated vendors publish intended trajectories to a shared schedule and negotiate conflicts peer to peer, with the explicit goal that no vendor changes its fleet manager; it has no stable wire protocol yet, and users report high processor load during negotiation [41]. The computation of a safety filter is measured in the runtime-assurance literature: on a spacecraft-docking problem, explicit optimisation cost 3.3 times explicit switching, and implicit filters that simulate the backup trajectory online 11–13 times [45]; the tutorial of Hobbs and colleagues lists computation time among the costs a safety layer must report [46]. Our safety program is explicit switching in the Simplex form [47, 48], chosen for that reason, and Section 5.1 reports its cost per call. Drone traffic management gives the only measured latency of a federated coordination service among operators: 189 ms on average, 95 % of exchanges within 531 ms, with operational impact traced to downed servers and misimplemented specifications rather than latency [66]. Decentralised auctions over lossy links have known degradation curves [42, 43, 44]; their message counts are reported per round, not per robot-hour, and never against a link budget.

2.4 Markets among owners who do not trust one another

When robots belong to different companies, an allocator is also a market, and the theory is unkind. For a single buyer and seller with private values, no incentive-compatible, individually rational mechanism is efficient without an outside subsidy [29]. The Vickrey–Clarke–Groves mechanism is truthful and efficient but runs a deficit, and with few bidders it is open to collusion by losing bidders and to shill identities [30, 31]. Trade-reduction double auctions buy truthfulness and budget safety by dropping the least valuable trade, which guarantees at least $1 - 1/q$ of the gains with q efficient trades, cheap in thick markets and expensive in a market of three owners [32]. When the mechanism can observe what an agent actually did, truthful mechanisms exist where unverifiable ones fail [33]; a rover's odometry, energy log and the shared trace make its costs checkable after the fact, which is the basis of the audits in Section 3.5. The one mechanism-design study of self-interested robots covers paths, not jobs, and finds that VCG needs an optimal solver, that approximate solvers lose strategyproofness unless maximal in range, and that three strategyproof, individually rational mechanisms work on dozens to hundreds of agents [34]. Freight carriers who collaborate save about 20–30 % in joint route planning, name “a reliable party to lead” and “a fair allocation of benefits” as the highest impediments [35], mostly share the gain by Shapley value [36, 37], and run request exchanges in five phases, selection, bundling, bidding, winner determination and profit sharing, under the constraint that no carrier's profit falls [38]; a survey of 120 such studies finds carrier alliances the most studied form and the managerial and technical issues of putting one into practice the least [65]. Ride-hailing data give the price of not coordinating: m operators raise the fleet by $(m - 1)\cdot D$, with D from 2.52 % in Manhattan to 67.14 % in Curitiba per extra operator, and with

50/35/15 shares in Singapore the smallest operator paid a 48.05 % fleet penalty against 8.48 % for the largest [39]; the theory predicts that unequal shares can make the excess grow linearly [40]. Our market experiment uses the same three-owner, 50/35/15 split, measures what each owner earns against working alone, reports the operator's budget rather than assuming it balanced, and tests one manipulation, bid scaling, with and without audits.

2.5 Our earlier work and the gap

Our first paper built a discrete-event simulation of cargo movement at a south-pole base and a pooled dispatch marketplace in which vehicles bid their marginal cost in hours [56]. The second added landing precision, pad standoff, a plume keep-out window and rover autonomy on NASA's 5 m terrain of the Connecting Ridge [57]. The third computed the base's Earth-link windows from ephemerides and terrain and priced their gaps [58]. The two companion papers of this series moved dispatch onto the robots over a terrain-limited mesh with learned bids [60] and gave the vehicles rules of the road enforced by a safety program checked three ways [61]. None of them said what a vehicle must carry to take part, or what owners are paid.

Table 1 sets the closest works against the features this paper needs. The pieces exist separately: a flight leader election and database for one owner, measured mesh behaviour underground, a negotiation protocol for mixed vendors on Earth, measured filter costs for one spacecraft, and mechanism design for carriers and cars. What we did not find is a sizing of the on-vehicle software and radio of a multi-owner lunar robot network against the vehicles that will fly, together with a measurement of its market's payments, fairness and manipulability in a logistics model with real link windows.

Table 1. Features of the closest prior work. “Yes” means the feature is part of the study's system, measurement or experiment; “–” means it is absent.

Work	Several owners	Flight or field hardware	Leader election, failover	Measured compute per decision	Measured radio traffic	Payments, individual rationality	Manipulation, audits	Lunar vehicles and terrain
CADRE [26, 27, 28]	–	Yes (flight)	Yes	–	–	–	–	Yes (one site)
ACHORD, CERBERUS [54, 55]	–	Yes (field)	–	–	Yes	–	–	–
Open-RMF [41]	Yes (vendors)	Yes (deployed)	–	– (reported as high)	–	–	–	–
Dunlap et al. 2021 [45]	–	–	–	Yes	–	–	–	–
NASA UTM, TCL4 [50, 66]	Yes	Yes (field)	–	–	Yes (latency)	–	–	–
Friedrich et al. 2024 [34]	Yes	–	–	–	–	Yes (paths)	Yes (strategyproof)	–
Freight collaboration [35, 36, 38]	Yes	–	–	–	–	Yes	–	–
Kondor et al. 2022 [39]	Yes	–	–	–	–	Yes (fleet penalty)	–	–
NASA LSR, LunaNet [15, 16]	–	Yes (concept)	–	–	Yes (coverage)	–	–	Yes
Our earlier work [56–58, 60, 61]	Yes	–	Yes (modelled)	–	Yes (messages per robot-hour)	–	–	Yes
This paper	Yes	– (scaled by a stated factor)	Yes (measured)	Yes	Yes (bytes, per type)	Yes	Yes	Yes

3. The vehicles, the network and the design

This section describes the vehicles of 2028–2032 as published, the radio network as it stands and as we model it, what Lunar ATC puts inside each vehicle, and how its marketplace works. Provenance follows the convention of [57, 60]: *Measured* values come from a cited measurement or document, *Spec* from a published specification or a company claim, *Derived* (or *Modelled*) from a computation on such values, and *Assumed* values are our choices, each swept in Section 4 or named in Section 7. Results of this study are never flagged; they come from the result files listed under Availability. Where the sources we reached hold no value, the text says so rather than supplying one.

3.1 The base and the shared engine

Everything in this paper runs on the engine of the companion papers [60, 61], which extends the discrete-event simulation of [56, 57, 58]. The base is on the Connecting Ridge between Shackleton and de Gerlache craters, on NASA Goddard's 5 m terrain tile (Measured). Its sites are the habitat and control centre (BASE), a cargo depot, a power plant, a construction site, an oxygen pilot plant, a science site and a permanently shadowed crater, with two landing pads 1 km from the base. Demand follows the landings and bulk flows of [56], scaled so that the two cargo-class vehicles are busy about 30 % of the time (load factor 2.806, Assumed, calibrated in [60]); landed cargo belongs to the owner who ordered it; a tenth of landed items are team items that need two vehicles (Assumed). The pooled marketplace of [56] offers every job to every vehicle in auction rounds, each vehicle bids the marginal cost in hours of adding the job to its plan, weighted by the item's urgency, and the best bid wins; [60] runs those rounds on the robots over the radio mesh of Section 3.3 and adds a learned correction to the bids; [61] adds the rules of the road and the safety program. With every bid multiplier at 1 the market run reproduces the pooled auction of [60] exactly, and the E48 timers leave the event trace bit-identical (Section 3.6).

3.2 The vehicles of 2028–2032 and what they carry

Table 2 lists the seven vehicle classes of the catalogue, read from the catalogue file so that every value, flag and source in the table is the one the simulation uses. Six of them make up the simulated fleet of eight vehicles owned by three companies: owner A has the crew rover CLV-1 and the cargo rover FLEX, owner B the crew rover Pegasus and the cargo rover Eagle, and owner C two small rovers and two hoppers (the fleet of [60]). The crew rovers are the two NASA contracted in May 2026 [1]; their payload of 80 kg and range of 10 km per charge are NASA's October 2026 statement [1], and their battery sizes are set so that the range comes out, because no battery size is published. The cargo rovers are the same companies' uncontracted designs [2, 3, 4]; FLEX's 2,000 kg is a company claim, and Eagle's 800 kg is the original Lunar Terrain Vehicle requirement [5] assumed for a vehicle whose capacity is not published. The small rover is the HL-MAPP of the same catalogue [3]. The hoppers are the MoonFall drones of 2028 [1, 7], with a propellant budget of 50 km over 14 days (Spec) and a cargo capacity we assumed, because they are survey drones; a smaller hopper, Micro-Nova, with 8–10 kg over 25 km [8], is in the catalogue and in none of the fleets simulated here.

Table 2. The vehicle classes of the catalogue (lunarsim/atc/vehicles-p5.json). Each cell gives the value the simulation uses and its flag: Spec (a published figure or company claim, with the research note's confidence High / Medium / Low), Modelled (derived from published figures by a stated rule, for example the modelled speed as 0.6 of the published top speed and the energy per kilometre from the traction model of [57]) or Assumed (no source; swept). The full source string of every value is in the file.

Class	Status	Owner	Mass	Payload	Top speed	Modelled speed	Energy	Battery	Works at night	Can join a team lift

Class	Status	Owner	Mass	Payload	Top speed	Modelled speed	Energy	Battery	Works at night	Can join a team lift
Astrolab CLV-1 (Phase-1 LTV)	contracted (NASA LTV Phase 1, 26 May 2026)	A	907 kg, Spec (High)	80 kg, Spec (High)	9.7 km/h, Spec (High)	5.8 km/h, Modelled (Low)	75.8 Wh/km, Modelled (Medium)	758 Wh, Assumed; 10 km, Spec (High) per charge	No, Assumed	Yes, Assumed
Lunar Outpost Pegasus (Phase-1 LTV)	contracted (NASA LTV Phase 1, 26 May 2026)	B	700 kg, Assumed	80 kg, Spec (High)	14.5 km/h, Spec (High)	8.7 km/h, Modelled (Low)	58.5 Wh/km, Modelled (Low)	585 Wh, Assumed; 10 km, Spec (High) per charge	No, Assumed	Yes, Assumed
Astrolab FLEX (cargo class)	not contracted	A	1,600 kg, Assumed	2,000 kg, Spec (High)	–	6.0 km/h, Assumed	133.7 Wh/km, Modelled (Low)	40,000 Wh, Assumed	No, Assumed	Yes, Assumed
Lunar Outpost Eagle (cargo class)	not contracted	B	1,500 kg, Assumed	800 kg, Assumed	–	6.0 km/h, Assumed	125.4 Wh/km, Modelled (Low)	40,000 Wh, Assumed	Yes, Spec (Medium)	Yes, Assumed
Lunar Outpost HL-MAPP (small cargo rover) × 2	company product	C	250 kg, Spec (High)	200 kg, Spec (High)	10.8 km/h, Spec (High)	6.5 km/h, Modelled (Low)	20.9 Wh/km, Modelled (Medium)	2,000 Wh, Assumed	No, Assumed	No, Assumed
NASA/JPL MoonFall hopper × 2	in development (Firefly delivery 2028)	C	250 kg, Spec (Medium)	10 kg, Assumed	–	20.0 km/h, Assumed	propellant: 50 km, Spec (Medium) in 14 d, Spec (Medium)	1,000 Wh, Assumed	No, Assumed	No, Assumed
Intuitive Machines Micro-Nova hopper	built (not deployed on IM-2)	– (in no fleet)	40 kg, Assumed	8 kg, Spec (High)	–	20.0 km/h, Assumed	propellant: 25 km, Spec (High) in 14 d, Assumed	500 Wh, Assumed	No, Assumed	No, Assumed

The capacity plateau. The largest single payload in the fleet is 2,000 kg, the largest pair that can couple lifts 2,800 kg together (the two-rover carry of [25] is the one measured case, at about 0.3 of the pair's combined mass), and the fleet's eight vehicles carry 3,380 kg in all if every one is loaded at once. Against that, NASA puts single base elements at 500–15,000 kg and occasional deliveries at 12,000–15,000 kg [5, 6], and the landers of the period put 150 kg (Blue Ghost [13]), 455 kg (Griffin as tested [12]), 1.5 t (Argonaut, from 2030–2031 [14]) and 3 t (Blue Moon MK1 [11]) on the surface at a time. Nothing heavier than 2,800 kg moves at all in this fleet, whatever the dispatcher does; the companion paper found that most of the value of sharing vehicles across owners is cargo that owners alone never deliver [60]. The plateau is a fact about the fleet, not about Lunar ATC, and we state it here because the sizing of Section 3.4 is for these vehicles and not for the cargo rovers NASA has not yet bought. Larger fleets in Section 4.2 multiply every vehicle class and the demand by the same factor.

3.3 The radio network

Earth. Earth is in view of the base when any part of its disc is above the terrain horizon of a 2.5 m antenna at BASE, the criterion of [58]; the windows are those of [58] for 2028–2031 and the same pipeline's windows to 2046 after that (Modelled). A message to Earth takes 2.6 s round trip plus 2.4 s of ground processing (Measured for the light time, Assumed for the processing, as in [56]). Earth in this paper never dispatches; it publishes prices and rules, receives the ledger, settles accounts and audits (Section 3.5).

LunaNet. The surface links of Lunar ATC are not LunaNet links. LunaNet 1.0 standardises the links between orbit and the surface and the network layer above any surface radio, and leaves the surface wireless link to

bilateral agreement [15]. Lunar ATC therefore assumes only what LunaNet provides to a surface user: IP routing, the Bundle Protocol for anything that can wait, and a shared time base from the navigation signal or the relay's clocks [15, 16] (Assumed that every vehicle holds it; a free-running oscillator would drift by about 86 ms a day at 1 ppm, Assumed, which the cooperative lifts of [60] cannot tolerate). The robot-to-robot link below IP is ours, and Section 3.4 states what it carries. Its physical layer is left open by this paper; the link budget of Section 4.2 (E48) is computed at one frequency, 2.4 GHz, the Wi-Fi-class radio the Griffin lander flies [18] (Provisional), with transmit powers of 1.0 W for the crew and cargo rovers, 0.25 W for the small rover, 0.5 W for the hopper and 5 W for the base mast (Assumed, no source). A 400 MHz UHF fallback would lose less at knife-edge diffraction over a low ridge [20], and a 390–405 MHz lunar band is in draft [24], but it is not budgeted here.

The mesh. Two robots can talk directly when their antennas see each other over the terrain and are within range. The mesh is the data product of [60]: line of sight for every pair of 2,421 points (the sites, the pads and samples every 100 m along every route) on the 5 m terrain with the Moon's curvature and no refraction, for antennas 2.5 m high on rovers and 1.0 m on small robots and hoppers (Assumed) and a 10 m mast at BASE that carries the base computer (Assumed, after the surface-relay concept of [16]). The range is 5 km and the clearance margin 0 m (Assumed). Line of sight is scarce on this ridge: two rover antennas within 500 m see each other in 50.5 % of point pairs and at 2–5 km in 9.3 %; for two 1 m antennas at 2–5 km the share is 6.3 %. The mast sees 23.3 % of the route points directly, none of them beyond 2 km from the base. Rovers nevertheless form one connected network from 2 km range upward; at 5 km 23.3 % of the points reach the mast in one hop, 58.1 % in two, and every point in at most four (median two); small robots leave four points with no neighbour at all within 2 km and need up to five hops. A message hop takes 10 ms (Assumed, terrestrial practice; no lunar measurement exists) and is lost with probability p, independently or in bursts of mean length 10 messages (Assumed), with p = 0.10 in the runs of [60] and in E48. E46 runs with p = 0 by design, a perfect radio (Section 4.2). A robot that nobody hears drives to a rally point that sees the mast (Assumed; a lost-link procedure every robot carries).

Why the numbers are what they are. On a smooth Moon the horizon of a 2.5 m antenna is 2.95 km and of a 10 m mast 5.89 km, so two rovers can never see each other beyond 5.9 km and a rover cannot see the mast beyond about 8.5 km [63]; the mesh above reaches less than half of that because the ridge is not smooth, which is the same finding as the relay viewsheds of [16]. A rover antenna cannot clear the first Fresnel zone at kilometre ranges (5.6 m at 2.4 GHz over 1 km, 13.7 m at 400 MHz), so a link in line of sight still sits near grazing; beyond a two-ray breakpoint of about 128 m at 2.4 GHz for 2 m antennas, loss grows about 40 log d instead of 20 log d; free-space loss alone is 100 dB at 1 km and 120 dB at 10 km at 2.4 GHz, and 85 and 105 dB at 400 MHz (all Derived in the research note from standard formulae [20, 63]). The lunar regolith's permittivity at the pole is unmeasured [21]. The link budget of E48 therefore reports a data rate at the worst hop under these models and not a measured range.

3.4 What Lunar ATC puts inside each vehicle

Lunar ATC is four pieces of software and the radio that carries their messages. Table 3 lists them per vehicle class, with the measurements E48 makes of each (Section 4.2). The design is ours; the parts of the vehicles themselves, their avionics, batteries and motor controllers, are not described here beyond what Section 3.2 needs, because few of their layouts are published.

The safety program. One function, `safety_check(state, proposal)`, is the only place where the rules of the road of [61] are decided. It answers two kinds of proposal. A *reservation* proposal asks whether this vehicle may hold a resource, a one-lane road segment, a junction tile or a pad apron, for a time window, and is decided

on the reservation entries the vehicle can see; a reservation that has not been confirmed is never assumed [49], so a lost message can only add delay. A *move* proposal is the planner's acceleration for the next step with the vehicle's kinematic situation, and is accepted only if, after the step, every rule still holds with the vehicle braking at its guaranteed deceleration; otherwise the first safe candidate, no acceleration and then full braking, is substituted. It is explicit switching in the Simplex form [47], a post-posed shield [48]: it never plans, it only refuses. The same function is called by the logistics model at job commit, by the encounter simulator at 10 Hz for every vehicle, and by the model checker for every move [61], so what E48 times is what was checked. Its state is the vehicle's own reservations, the broadcasts it has heard and the published constraints: landing closures, crew zones, the priority ladder.

The planner. The planner may be anything; the safety program does not trust it. In this study it is the nominal driving logic of [57] for moves and, for bids, the marginal-cost bid of the marketplace [56] with the learned correction of [60]: a linear model of 13 weights, which was the model that improved on hand-built bids, or a multilayer network of 1,537 parameters, which did not. Both are read from the policy files; both are small enough that the question for sizing is the auction's bookkeeping, not the arithmetic of a bid.

The radio unit. The unit floods bids, winners' rebroadcasts, status reports, reservation requests and grants, team-lift proposals, answers and commits, and ledger entries over the mesh, hop by hop through other robots or the mast, with the loss and latency of Section 3.3. A robot in a part of the network that holds the mast knows every waiting item; a robot elsewhere knows only the items at sites where one of its part's robots stands [60]. Status and intent are broadcast at 1 Hz for conformance [61]. E48 does not serialise these messages; it counts the traffic from each run's events. Every road traversal produces one reservation request, one grant and one release, every auction round its bid, award and claim, every landing one announcement, and every robot one status message a second (Assumed rate, after rule R9 of [61]); each type has an assumed byte layout, ids, times and positions as 32-bit values under an 8 B header (Assumed; no serialiser is flown). The winners' rebroadcasts, the team-lift proposals, answers and commits, and the ledger entries are not in that traffic; the ledger is sized separately from the year's jobs. The design targets we set before measuring were 100 B/s per robot, a 16 kbit/s-class link for 20 robots, and under 1 MB a day of market messages (Assumed; hypothesis H10 of [59]). For comparison, the one measured inter-rover coordination load on record, a two-rover convoy controller, used 13.8 kB/s [25].

Local copies. Each vehicle keeps a copy of the reservation entries it has heard and of the ledger entries it has produced or relayed, so that it can check a proposal while out of contact (rule R7 of [61]: a vehicle out of contact keeps the reservations it holds and takes no new ones) and can give Earth its own account of every job. E48 measures the peak memory of the on-robot part, planner, safety program, reservation table and ledger copy together, in our implementation.

The base computer and its spare. The base computer on the 10 m mast holds the shared reservation table, the manifests of waiting items, the market's rounds when the leader regime runs them, and the ledger that Earth settles from. It has a spare board; in the model the computer fails as a process with a mean time between failures of 2,000 h and a repair time of 24 h (both Assumed, swept in E48). While it is down the robots elect a leader among themselves, the robot that reaches the most others over the mesh, after a re-election delay of 0.5 h (Assumed, after CADRE's wake cycle [27]; E48 does not measure this delay, it copies the assumed value and measures the share of time without a leader at it); robots outside the leader's part of the network act on their last instructions and, when idle, drive to the rally point. The leader runs the market for the robots it reaches; the reservation table it holds is the union of what those robots have heard. When the base computer returns after repair, or two parts of the network merge, the base computer's state is rebuilt from the robots: what merges is the

assignment table, by the rule that an earlier commit keeps its reservation and a job is never assigned twice [61]. E48 checks after every rebuild that no job is assigned twice. The ledger itself is not simulated; its size is estimated from the year's jobs (Section 4.2). The auction regime of [60] needs no leader: nothing waits for one, and the base computer's failure only removes the mast from the mesh.

Table 3. What Lunar ATC puts inside each vehicle class (our design) and what E48 measures of each. The measured values are in Table 5; this table states only what is there and how each value is obtained. “Design” = our choice, stated here; “Measured” = timed or counted on our implementation; “Derived” = computed from measured values and a stated factor; “Assumed” = a value with no source, swept.

Class	Software parts (Design)	Compute	Memory	Radio	Power	If it fails
Crew rover (Phase-1 LTV: CLV-1, Pegasus)	Safety program (moves and reservations); planner with learned bids; radio unit; local reservation table and ledger copy; conformance broadcast at 1 Hz	Measured: CPU per safety check at 10 Hz and per auction round; Derived: scaled to the reference flight processor	Measured: peak of the on-robot part	Antenna 2.5 m (Assumed); Derived: rate at the worst hop, latency budget, share of route points in direct view of the mast; Measured: bytes per message type, messages per robot-hour	Assumed, no published source: a GR740-class board plus the radio unit	Keeps held reservations, takes no new ones, stops at the next safe point when they expire; rallies to a site that sees the mast when idle and unheard
Cargo rover (FLEX, Eagle)	As the crew rover; loaded vehicles rank higher in the priority ladder	As above	As above	Antenna 2.5 m; as above	As above	As above
Small rover (HL-MAPP)	As the crew rover, without team lifts	As above	As above	Antenna 1.0 m (Assumed): the class most often out of line of sight (Section 3.3)	As above	As above
Hopper (MoonFall)	Safety program for landing closures only (it flies over roads and road rules); planner; radio unit; ledger copy	As above, fewer rules	As above	Antenna 1.0 m in flight and on the ground (Assumed)	As above	As above; a hopper on the ground is parked in the ladder
Lander	The Lunar ATC radio unit and the landing-announcement part only, with no planner and no safety program (change 7 of [59]); not sized here	–	–	Its own Earth link is for flight and is not part of Lunar ATC; the Lunar ATC radio unit on the lander broadcasts the landing announcement on the mesh, which is the landing traffic E48 counts; Earth also holds the landing time	–	A late or waved-off landing extends the closure [61]
Base computer (on the mast) and spare	Shared reservation table; manifests; the market's rounds under the leader regime; ledger; Earth gateway	Measured: CPU per auction round at one, four and eight times the fleet	Estimated: ledger size per year from the year's jobs; the table	Mast 10 m (Assumed); the Earth link of [58]	As above; the relay concept of [16] draws 131 W for its communications (Measured, for a different radio)	Spare board; MTBF 2,000 h, repair 24 h (Assumed, swept); robots elect a leader after 0.5 h (Assumed); the assignment table is rebuilt from the robots on return (checked: no job assigned twice)

3.5 The marketplace

Jobs and bids. The base computer posts every waiting item with its mass, origin, destination and urgency. In each auction round every free vehicle computes its bid, the marginal cost in hours of inserting the job into its plan, weighted by the item's urgency [56], adjusted by the learned correction [60], and floods it; the lowest bid wins, ties going to the lower vehicle number. Under the leader regime the base computer or the elected leader collects the bids; under the auction regime the robots settle the winner among themselves by the rebroadcast rule of [42, 43, 44]. A team item is won by a pair that agrees by a two-phase commit over the radio, with a

rendezvous timeout of 1 h (Assumed) [60]. All of this is the companion paper's machinery; what this paper adds is the money.

Private costs. Each owner has a private cost of running a vehicle: energy, wear and the opportunity cost of the vehicle's time, drawn per owner from an assumed distribution: the engine's job cost, wear plus operator time, times a multiplier $\exp(N(0, \sigma))$ with $\sigma = 0.2$ (Assumed), one draw per owner per seed, the same in every run of that seed; a range of 0.1–0.3 is declared and not swept in E46, because a sweep needs new physical runs (Section 7). A bid is a vehicle's cost in hours; its owner's money cost is that time at the owner's private rate. An honest owner bids its true marginal cost.

Four payment rules. E46 runs the pooled market under four rules. A rule prices a run after it has been simulated: the auction allocates the jobs on the asks (honest, or scaled for the one deviating owner), and the rule then decides what is paid for the jobs so allocated, so with honest asks the four rules share one allocation. *None*: no money changes hands; each owner's vehicles work for everyone and the owner bears its own costs, the cooperative case of [60]. *First price*: the carrier of each job is paid its own ask, and the shipper of each item pays the posted tariff times the item's tonne-kilometres; it is the rule under which an owner gains by asking above its cost, which H5's first clause tests. *Per-item second price*: the carrier of each job is paid the larger of its own ask and the lowest ask a competing free vehicle would have made for the job at the dispatch instant (that vehicle's cost times its owner's private multiplier, and times the scaling factor if the vehicle belongs to the deviating owner, so a deviator's own other vehicle competes at its scaled ask). That alternative is reconstructed from the run, not bid, because the auction collects no second bid; a team lift is paid the sum of the two lowest such alternatives, split equally between the two carriers, each floored at its own ask; when fewer competing vehicles than the job needs could have carried the item, the tariff times the item's tonne-kilometres is the reserve; the shipper pays the tariff. It is Vickrey's rule [30] applied item by item, and because the auction is sequential across items and rounds it is not strategyproof across rounds, which the manipulation test measures. The shares of jobs with a competing alternative and priced by the reserve are reported in Section 5.5. *Posted price per tonne-kilometre with Shapley sharing*: Earth publishes a tariff per tonne-kilometre before the year; the carrier of a delivered item is paid the tariff times the item's mass and road distance and the item's shipper pays the same, so the operator's budget is balanced by construction; at the end of the year the coalition's value, its true carrier costs plus its cargo's delay costs with the sign reversed, is shared among its members by their Shapley values [37] through transfers, computed exactly from the seven non-empty coalitions' values, which five runs per seed give (the owners-alone run supplies the three singletons, three pair runs the pairs, and the pooled run the grand coalition), the rule that freight alliances most often use [36]. In E46 the posted price is 19,828.85 USD per tonne-kilometre with equal shares and 19,908.58 USD per tonne-kilometre with 50/35/15 shares, set before the main runs as the price at which the carriers' true costs are recovered over ten calibration seeds (101–110) that the main runs do not use; the audit penalty multiple is $m = 3$ (Assumed; the design file declares 2–5 and E46 runs one value).

Money flow and settlement. No money moves on the Moon. The base computer's ledger records every posting, bid, grant, delivery and payment due, signed by the vehicle that produced the entry (Design; the signing is not simulated). Earth receives the ledger in every window and settles afterwards: each owner's cash, what its vehicles were paid as carriers less the tariffs it paid as a shipper and any fines, is netted across owners, and the operator's budget, its receipts (the tariffs shippers pay plus fines) less its payouts to carriers, is reported per year as a surplus or a deficit and never assumed balanced [29]. Under the posted price the Shapley transfers of the year's end are settled on Earth with the cash. The amounts settled on Earth and left open at the end of the year (pay due on jobs unfinished at the horizon) are outcomes of E46.

Individual rationality. An owner is individually rational in the market if its profit over the year, earnings less its private costs, the tariff it pays for its own cargo and its cargo's delay cost, is at least what it would have made working alone with its own vehicles on its own cargo: the owners-alone regime of [60] decided on the Moon (A_MOON of the companion paper [60]) on the same seeds, so that the comparison carries no Earth-availability effect (change 7 of [59]). The alone profit is dominated by the owner's cargo's delay cost at its own value of time, because an owner alone has few vehicles for its cargo and no one pays it. E46 reports each owner's mean profit under each rule against its mean alone profit and counts the owners, per cell, whose market profit fell below alone; under the posted price the quantity compared is the owner's Shapley value of the coalition's value, and H5's third clause tests it with a bootstrap interval for every owner in every coalition. As the harder comparison, E46 also reports each owner's profit under each paid rule against its profit in the pooled market with no payments, paired by seed (Section 5.6), which asks whether a payment rule leaves an owner worse off than the cooperative market does.

Manipulation. One owner scales every bid of its vehicles by a factor k in {0.8, 0.9, 1.1, 1.25} while the other two bid honestly (Assumed levels, from the research design); the test is run for each owner in turn. The outcome is the largest gain in that owner's profit over honest bidding across the four factors, the factor that gives it, and the cost to the system: the change in the mean wait of urgent cargo, the primary outcome of [60], on the same seeds.

Audits. Because every job is in the ledger with the vehicle's reported cost and its logged energy and distance, Earth can audit afterwards. The model charges the expected penalty of an audit rather than drawing one. At an audit share q in {0, 5, 20 %}, every job of an owner that asks under first or second price is charged q times m times the absolute difference between its ask and its true cost, so asks above and below cost are fined alike and no random draw enters the paired comparison; the multiple m is stated with the tariff above. The outcome is what the audits recover of the manipulating owner's gain, and whether the gain net of expected penalties is still positive at each q.

3.6 Checks

The engine's existing checks are those of [56, 57, 58, 60, 61]. This paper's own checks are part of the phase-5 suite: with every bid multiplier at 1, the market run reproduces the pooled auction exactly (A8); with one owner there are no payments between owners under any rule; the operator's budget is zero under no payments and the posted price, and under first and second price it equals the tariff receipts less the payouts, all of it between that owner and Earth (A8); the Shapley payoffs sum to the grand coalition's value on every seed (A8); asks scaled by k = 1.0 reproduce the honest payments exactly (A8); the audit at q = 0 charges nothing and at q = 1 charges m times every misreport (A8); across base-computer failures and recoveries and across mesh partitions, no cargo item is ever assigned to two jobs (the ledger itself is sized, not simulated) (A8); the E48 timers leave a run's event trace bit-identical to a run with timing off (X7) (seven checks in 9 rows; outcomes in Section 5.8). In addition, 49 further checks of the engine's phase-5 verification suite (`lunarsim/vv5.py`, results in `results/vv_phase5.json` and on the live site's verification page) bear on this paper: 46 verifications that the code does what the model says, one validation against references outside the model, two calibration reproductions, by Sargent's technique analytic (26), traces (8), comparison to other models (7), extreme condition (4), metamorphic testing (3), historical data (1). Deterministic identities are held to 10^{-9} relative or exact; stochastic checks pass when the error is within a tolerance set in advance, and a 99 % batch-means interval that misses the reference is recorded as a bias flag, not a failure; the checks ran at commit `e63616c`. Their outcomes are in Section 5.8.

4. Experiment design

4.1 Questions and hypotheses

The pre-registration [59] states two hypotheses for this paper. **H5 (incentives).** First-price bids let at least one owner gain by shading; per-item second price reduces the largest gain; posted price with Shapley sharing leaves no owner worse off than alone in any tested coalition; the operator's budget is reported, not assumed balanced. **H10 (sizing, exploratory).** The safety check, the bidding and the radio traffic of Lunar ATC fit a flight-class computer, reported as measured CPU time, memory and bytes per robot-hour scaled to a reference radiation-tolerant processor with a stated factor, and a 16 kbit/s-class link for 20 robots. H5 was written before any run as part of the original plan and was moved to this paper on 27 October 2026, when the three papers were regrouped (change 1 of [59]). H10 was added on the same date, before any run of either experiment. Two things must be said about H5 as registered. First, the reduced design of 27 October had no first-price rule, which would have left H5's first two clauses untestable; a first-price cell was restored on 29 October (change 7 of [59]), before any E46 row, so H5 is tested as registered. Second, H5 carried no numeric decision rule. The rules, registered as change 7 of [59] before the first E46 row, are: clause 1 is supported if, under first price, the largest gain in an owner's annual profit over bidding truthfully across the tested k has a 95 % interval above zero; clause 2 if the largest such gain under per-item second price is smaller than under first price with the interval of the paired difference excluding zero; clause 3 if, for every owner in every tested coalition, the 95 % interval of the owner's profit in the coalition less its profit alone does not lie entirely below zero, with the point estimates reported either way; intervals are percentile bootstraps over seeds with 20,000 resamples, and the operator's budget is reported per rule, not tested. H10 is exploratory and has no decision rule; it is reported as the measured values against the stated capacities.

4.2 Experiments

E46, the market. Three owners, with equal shares of the cargo and with 50/35/15 shares (the split of [39]), at the middle load, under the four payment rules of Section 3.5; for the two bid rules, first price and second price, the manipulation test with k in {0.8, 0.9, 1.1, 1.25} for each owner in turn, and the audit share q in {0, 5, 20 %}; the owners-alone regime on the same seeds as the individual-rationality baseline, and every subset of owners for the Shapley shares. Seeds 1–30 in every cell, a simulated year each. 1,080 physical runs, 30 seeds by 2 splits by 18 run kinds (the pooled market, the oracle, the owners alone, the three pair coalitions and one owner at a time scaling its asks by each of the 4 factors); payments are computed afterwards from each run's job log, so the four rules and the 3 audit shares share one allocation per seed and cost no further runs, giving 152 cells of 30 seeds. The first-price cell is the one restored by change 7 of [59]. The dispatcher is the pooled market run by the base computer or leader over the mesh of [60]. E46 sets the message loss to 0.00 in every cell, the owners-alone and coalition runs included: a perfect radio by design, so that paired differences show incentives and not message-loss noise, which the companion paper [60] covers. The road layer of [61] is on with its defaults.

E48, sizing. Our implementation is instrumented and run, not changed: processor time per `safety_check` call and per auction round, as the median and the 99th percentile over every call in a run; peak memory of the on-robot part; and the radio traffic computed from each run's event counts (road traversals, auction rounds, landings) at the assumed byte layouts of Section 3.4, plus a status message at 1 Hz per robot, from runs at one, four and eight times today's fleet and demand, the density ladder of [61]; the ledger's size is the year's jobs times the entry's bytes. The instrumented runs last 365 days at 1 times today's fleet, 90 days at 4 times today's fleet, 60 days at 8 times today's fleet; the ledger's yearly growth at four and eight times is extrapolated from jobs per

hour. The measurements are scaled to a reference radiation-tolerant flight processor by the ratio of its published benchmark to this laptop's, flagged Assumed with a range. The reference flight processor is the Frontgrade Gaisler GR740 (rad-hard quad-core LEON4FT, 250 MHz), at 492.5 CoreMark per core (source: Cobham Gaisler, GR740 benchmark results, ESA DSP Day 2016: 1.97 CoreMark/MHz/core x 250 MHz) with 256 MB of memory (Assumed: SDRAM of a typical GR740 single-board computer); the laptop is an Intel Core i5-10500H (6 cores, 2.5 GHz base, 4.5 GHz turbo) at 27,000 CoreMark per core, itself assumed (Assumed: 4.5 GHz turbo (Intel spec) x 6.0 CoreMark/MHz (typical for this core generation; no published CoreMark for this chip found), range 4-7 CoreMark/MHz); the factor is their ratio, 54.8, with the range 36.5–64.0 from the laptop's CoreMark range, and the CPU share uses the 99th-percentile call time at every call, so it is an upper bound (Assumed). The scaled values are compared with the design targets of Section 3.4. The link budget per vehicle class takes the antenna height, frequency and transmit power of Section 3.3 at the mesh's worst hop, and reports the data rate at that hop, the latency of a three-hop exchange of a 100 B message at the hop latency of Section 3.3 (one hop for the base computer) plus one scaled safety check, against a decision interval of 1 s (Assumed; a target of this paper's design that no design file holds), and the share of route sample points in direct view of the mast. Failover sweeps the base computer's mean time between failures. The failover sweep runs the base computer at an MTBF of 500 h, 2,000 h, 8,000 h, a repair time of 24 h and a re-election delay of 0.5 h, on five seeds of C_LEADER, message loss 0.1, 30 % load, 1x, one year, flagged Assumed (B-72 defaults; sweep); the re-election delay is an input, not a measurement. The sweep measures the base computer's downtime share and the share of time without a leader, and checks the assignment table after every rebuild for a job assigned twice. 21 runs were instrumented: six timing runs, seeds 1, 2 at 1, 4, 8 times today's fleet (C_AUCTION, message loss 0.1, roads atc (15 % single-lane), dispatch to a fixed point), and 15 failover runs, seeds 1–5 at the three MTBF levels (C_LEADER, message loss 0.1, 30 % load, 1x, one year); the auction round and the peak memory are measured once per run, not per class.

4.3 Seeds and common random numbers

Every cell of E46 uses seeds 1–30, so a seed's demand, landings and faults (and message losses, of which E46 has none) are identical across payment rules, share splits, scaling factors and audit shares, and the difference between two cells on one seed is the effect of the rule alone; the owners-alone and coalition runs of a seed share its demand too. Pairs are formed on the seed. E48 instruments existing runs and adds no randomness of its own; where it times a fresh run it uses the seeds of the run it reproduces. No run is excluded after the fact; a run that crashes is fixed and repeated with the same seed, and the crash is reported in Section 5.8.

4.4 Outcomes

The outcomes of E46 are, per owner and year: profit under each rule; the difference between the owner's Shapley value in each coalition and its alone profit on the same seed (individual rationality); the operator's budget surplus or deficit; the money settled on Earth and left open at the year's end; the efficiency loss against the oracle of [60] in system cost, true carrier cost plus delay cost; and, for the manipulation cells, the largest gain from scaling, its factor, its cost to the system in cargo delay hours and in system cost, and what each audit share recovers of it. The outcomes of E48 are the measurements of Section 4.2 per vehicle class, the scaled values, the link-budget quantities per class (the rate at the worst hop, the latency of a decision, the share of route sample points in direct view of the mast), the base computer's downtime share and the share of time without a leader per MTBF level at the assumed re-election delay, and the count of jobs found assigned twice after a rebuild of the assignment table, which must be zero.

4.5 Statistics

Differences between cells are paired means over seeds with a percentile bootstrap 95 % confidence interval of 20,000 resamples [62]. H5's clauses use the one-sided rules of Section 4.1; the Holm correction of the companion paper's confirmatory family does not apply here, because H5 is a single hypothesis with three clauses reported separately. Sizing values are reported as medians and 99th percentiles over calls with their counts, scaled values as a point estimate with the range the scaling factor's range implies, and the H10 comparison as the ratio of each scaled value to its capacity. All intervals are descriptive of the simulated population of seeds, not of a lunar fleet. The analyses are descriptive (cell means), inferential (paired bootstrap intervals), causal by design (same seeds, one factor changed) and, for the link budget, mechanistic.

4.6 Choices made after looking at results

Two choices were made before either experiment ran and are recorded here. First, the market experiment was reduced on 27 October 2026, when the three papers were regrouped: the McAfee trade-reduction rule, charger hogging, false availability and cherry-picking of easy items were dropped, and the first-price rule with them; the first-price rule was restored on 29 October (change 7 of [59]) because H5 needs it, leaving four rules, one manipulation and the audits; the reduction was made for the time the runs would take, before any E46 row existed, and its consequence for H5 is stated in Section 4.1. Second, the dispatch fixed-point fix of the companion papers (change 5 of [59]) was adopted before any run of E46 or E48, so neither experiment has a before-and-after. One choice was made after the runs. The decomposition of the largest first-price gain into the part the scaled asks pay with the allocation held fixed and the change in the deviator's own cargo delay once its asks move jobs was made after E46 had run, at the lead's request, when the registered interval was seen to cover zero; it is reported in Sections 5.7 and 6.1, in the abstract and in the conclusion, labelled post hoc each time, and no registered verdict depends on it. Dated changes of [59] that touch this paper: Change 7, 29 Oct 2026 (before any E46 or E48 row; recorded while drafting P3 draft 0): H5's design and rules; Change 10, 30 Oct 2026 (after every experiment through E48; no experiment rerun): cargo ids were not unique; fixed, outcome-neutral. No other choice was made after looking at data.

4.7 Compute

Runs are single-process on one laptop shared with other work, with a per-run limit of five minutes, as in [60, 61]. A simulated year of E46 took 7 s on average and the experiment 2.1 h of processor time. The instrumented runs of E48 took 63 s each on average and 0.1 h of processor time in all.

5. Results

Sections 5.1 to 5.4 give E48: processor time and memory per vehicle class against the flight-processor capacities (H10), the radio traffic and the ledger, the link budget per class, and failover. Sections 5.5 to 5.7 give E46: the four payment rules, individual rationality per owner, and manipulation with audits (H5). Section 5.8 lists what did not work. Table 5 holds every sizing value, Table 6 the link budget, Table 7 every cell mean of E46 and Table 8 every owner's Shapley value in every coalition against working alone; the text describes them in that order. Intervals are 95 % bootstrap intervals over seeds unless stated. Nothing in this section is interpreted; Section 6 does that.

5.1 Processor time and memory per vehicle class (H10)

From the catalogue, the fleet's largest single payload is 2,000 kg and its largest team lift 2,800 kg (Section 3.2); the sizing below is for that fleet. Measured on Intel Core i5-10500H (6 cores, 2.5 GHz base, 4.5 GHz turbo) and scaled to the Frontgrade Gaisler GR740 (rad-hard quad-core LEON4FT, 250 MHz) by a factor of 54.8 (range 36.5–64.0, Assumed): across the four robot classes a safety check took 577 µs–583 µs at the median and 693 µs–763 µs at the 99th percentile over 20,000 calls per class, worst on the crew rover; an auction round took 1.98 ms at the median and 20.5 ms at the 99th percentile over 84,367 rounds; the on-robot part peaked at 0.39 MB. The base computer's reservation check over the shared table, a different function from the vehicles' safety check, took 136 µs at the median and 191 µs at the 99th percentile, and its auction round 1.98 ms and 20.5 ms (Table 5). H10, exploratory, holds on every count: the safety check at 41.8 % of one core of the reference processor at the worst class (the crew rover), the on-robot part at 0.4 MB of 256 MB, and the radio traffic at 7.68 kbit/s for 20 robots against the 16 kbit/s target, and the market's messages at 0.005 MB a day against the 1 MB target, with the scaling factor at 54.8 (range 36.5–64.0); the bidding, which H10 also names, has no registered capacity (Section 5.1 reports the scaled round against the one-second decision). Scaled by the same factor, an auction round on the crew rover takes 0.11 s at the median and 1.12 s at the 99th percentile, above the one-second decision that the link model of Section 5.3 assumes.

Table 5. Sizing per vehicle class, measured on our implementation (Intel Core i5-10500H (6 cores, 2.5 GHz base, 4.5 GHz turbo)) and, where stated, scaled to the Frontgrade Gaisler GR740 (rad-hard quad-core LEON4FT, 250 MHz) by 54.8 (Assumed, range 36.5–64.0). CPU per safety check in microseconds and per auction round in milliseconds, median / 99th percentile; peak memory of the on-robot part in MB; power with its flag. The auction round and the peak memory are measured once per run and repeated in every row; the base computer's timed check is its reservation check over the shared table.

Class	Safety check, base: reservation check (µs)	Auction round (ms)	Peak memory (MB)	Power (W)
Crew rover	583.4 / 763.3	1.98 / 20.5	0.4	12.0, Assumed
Cargo rover	578.2 / 731.1	1.98 / 20.5	0.4	12.0, Assumed
Small rover	579.9 / 738.8	1.98 / 20.5	0.4	6.0, Assumed
Hopper	577.0 / 693.4	1.98 / 20.5	0.4	8.0, Assumed
Base computer	136.2 / 190.8	1.98 / 20.5	0.4	25.0, Assumed

5.2 Messages and the ledger

With the assumed byte layouts of Section 3.4 the message types weigh: status 48 B; reservation_request 32 B; grant 32 B; release 16 B; bid 40 B; award 32 B; claim 32 B; landing_announcement 64 B; ledger_entry 128 B. Computed from each run's event counts and the status rate, the fleet generated 3,600.3 messages and 172,812 B per robot-hour at today's fleet, 3,604.2 and 172,945 B at four times, and 3,613.9 and 173,280 B at eight times; at today's fleet that is 48.0 B/s per robot against the 100 B/s target and 7.68 kbit/s for 20 robots against 16 kbit/s. The ledger grows by 0.3 MB a year at today's fleet, 1.4 MB at four times and 2.7 MB at eight, extrapolated from each run's jobs per hour. The market's messages come to 0.005 MB a day at today's fleet. The status message at the assumed 1 Hz is 99.99 % of the messages and 99.99 % of the bytes at 1 times, 99.88 % of the messages and 99.92 % of the bytes at 4 times, 99.62 % of the messages and 99.72 % of the bytes at 8 times; the traffic counts reservation requests, grants and releases from road traversals, bids, awards and claims from auction rounds, and landing announcements, and omits rebroadcasts, team-lift proposals, answers and commits, and ledger entries.

5.3 The link budget per vehicle class

Per class, under the models of Section 3.3, at 2.4 GHz with the data rate at the worst hop, the modelled latency of a 100-byte message (10 ms per hop plus transmit time plus the scaled safety check, against the one-second decision the model assumes) and the share of route samples in direct view of the mast: the crew rover (antenna

2.5 m, 1.00 W): 6,976 kbit/s at the worst hop, 72 ms over three hops, direct view of the mast from 23.3 % of the route samples; the cargo rover (antenna 2.5 m, 1.00 W): 6,976 kbit/s at the worst hop, 70 ms over three hops, direct view of the mast from 23.3 % of the route samples; the small rover (antenna 1.0 m, 0.25 W): 5,010 kbit/s at the worst hop, 71 ms over three hops, direct view of the mast from 19.2 % of the route samples; the hopper (antenna 1.0 m, 0.50 W): 5,988 kbit/s at the worst hop, 68 ms over three hops, direct view of the mast from 19.2 % of the route samples; the base computer (antenna 10.0 m, 5.00 W): 9,289 kbit/s at the worst hop, 20 ms over one hop, direct view of the mast from 100.0 % of the route samples. The 1 s decision interval of the latency budget is the Assumed target of Section 4.2, and the latency reported against it is for a move decision, a three-hop exchange plus one safety check, not for a market round; the auction round's time against the same interval is in Section 5.1.

Table 6. Link budget per vehicle class under the models of Section 3.3: antenna height (m), frequency (GHz), transmit power (W), data rate at the worst hop (kbit/s), modelled latency of a 100-byte message over three hops (one for the base) plus the scaled safety check (ms), and the share of the route samples in direct view of the base mast within 5 km (multi-hop reach is higher and not measured).

Class	Antenna (m)	Frequency (GHz)	Power (W)	Worst hop (kbit/s)	Modelled latency (ms)	Direct view of mast (share of route samples)
Crew rover	2.5	2.4	1.00	6,976.4	72	23.3 %
Cargo rover	2.5	2.4	1.00	6,976.4	70	23.3 %
Small rover	1.0	2.4	0.25	5,010.4	71	19.2 %
Hopper	1.0	2.4	0.50	5,987.8	68	19.2 %
Base computer	10.0	2.4	5.00	9,289.1	20	100.0 %

5.4 Failover

At an MTBF of 500 h the base computer was down 4.21 % of the time, the fleet had no leader 0.09 % of the time with the assumed re-election delay of 0.5 h, and the base computer's assignment table was rebuilt after repair 15.4 times a year on average over five seeds; at an MTBF of 2,000 h the base computer was down 1.10 % of the time, the fleet had no leader 0.02 % of the time with the assumed re-election delay of 0.5 h, and the base computer's assignment table was rebuilt after repair 4.0 times a year on average over five seeds; at an MTBF of 8,000 h the base computer was down 0.38 % of the time, the fleet had no leader 0.01 % of the time with the assumed re-election delay of 0.5 h, and the base computer's assignment table was rebuilt after repair 1.4 times a year on average over five seeds. After every rebuild the table held no job assigned twice.

5.5 The four payment rules

With honest bids the four rules share one allocation, so the items delivered and the cargo delay are the same under every rule on every seed (the paired differences between rules in delay and in deliveries are exactly zero), and the market's system cost, true carrier cost plus delay cost, was above the oracle's by 12.7 % with equal shares and 35.9 % with 50/35/15 shares under every rule. Under no payments no money moved; under the posted price the budget was balanced by construction, the same tariff paid in by shippers and out to carriers. Across the three paid rules and both splits the cash settled on Earth was 6.7 M USD to 20.9 M USD a year and the balance still open at the year's end at most 0.01 M USD (Table 7). Under the per-item second price, with equal shares 16 % of jobs had a competing free vehicle, 84 % were priced by the tariff reserve and 64 % paid the carrier its own ask; with 50/35/15 shares 15 % of jobs had a competing free vehicle, 85 % were priced by the tariff reserve and 63 % paid the carrier its own ask. The operator's budget with honest asks, receipts less payouts with its 95 % interval over seeds, was −2.50 M USD [−4.78, −0.22] under first price with equal shares; −2.76 M USD [−5.08, −0.44] under first price with 50/35/15 shares; −12.71 M USD [−14.36, −11.11] under the per-item second price with equal shares; −13.07 M USD [−14.80, −11.38] under the per-item second price with 50/35/15 shares; zero under

the posted price and under no payments by construction. Under first price with honest asks the carriers are paid their true costs, so that budget is the tariff receipts (45.0 M USD a year with equal shares) less the true carrier cost (47.5 M USD); the tariff was set to recover that cost on the calibration seeds (Section 3.5). The Shapley sharing is settled by transfers between owners, Shapley payoff less the owner's cash profit under the posted price on the same seed: with equal shares owner A +112 M USD [96, 129], owner B +88 M USD [76, 100], owner C −200 M USD [−224, −176]; with 50/35/15 shares owner A +34 M USD [22, 46], owner B +44 M USD [29, 62], owner C −78 M USD [−94, −62]. With the transfers included, the cash settled on Earth under the posted price is 214.4 M USD [190.9, 238.4] a year with equal shares and 90.3 M USD [77.1, 105.7] a year with 50/35/15 shares, against the tariff cash alone in Table 7.

Table 7. Cell means over the seeds of E46 per payment rule and share split, honest bids, no audits, in M USD a year. Profit is each owner's earnings less its private costs, the tariff it paid for its own cargo and its cargo's delay cost; under the posted price it is instead the owner's Shapley value of the coalition's value (carrier costs plus delay costs, sign reversed), the quantity H5's third clause compares with alone. Budget is the operator's surplus (+) or deficit (−). Efficiency loss is the market's system cost (true carrier cost plus delay cost) above the oracle of [60], the same under every rule because the rules share one allocation. Settled is half the sum of the owners' absolute net cash flows; open is the pay due on jobs unfinished at the year's end. IR violations count the owners whose mean market profit was below their mean alone profit.

Rule	Split	Runs	Profit A	Profit B	Profit C	Budget	Efficiency loss	Settled on Earth	Open at year end	IR violations
None	equal shares	30	−54.2	−53.7	−37.2	0.00	12.7 %	0.0	0.000	0
None	50/35/15 shares	30	−61.3	−59.9	−15.0	0.00	35.9 %	0.0	0.000	0
First price	equal shares	30	−47.7	−43.0	−52.0	−2.50	12.7 %	16.1	0.004	0
First price	50/35/15 shares	30	−62.3	−49.8	−21.4	−2.76	35.9 %	10.1	0.006	0
Second price (per item)	equal shares	30	−40.8	−39.8	−51.8	−12.71	12.7 %	20.9	0.006	0
Second price (per item)	50/35/15 shares	30	−55.4	−46.5	−21.2	−13.07	35.9 %	12.8	0.009	0
Posted price, Shapley sharing	equal shares	30	+65.6	+41.0	−251.7	0.00	12.7 %	14.7	0.005	0
Posted price, Shapley sharing	50/35/15 shares	30	−27.0	−9.7	−99.6	0.00	35.9 %	6.7	0.009	0

5.6 Each owner against working alone

Working alone on their own cargo on the same seeds, the owners' mean profits were −106.7 M USD, −118.7 M USD and −723.4 M USD a year for A, B and C with equal shares and −141.9 M USD, −119.0 M USD and −313.0 M USD with 50/35/15 shares (alone profit is the negative of the owner's vehicle costs and its cargo's delay cost, with no receipts). Under every rule and split each owner's mean market profit (Table 7) was above its alone profit: the market left an owner below alone in 0 of the 24 owner-rule-split cases with honest bids and in 0 of the 456 owner-cell cases over every cell. Under the posted price the profit of Table 7 is the owner's Shapley value of the coalition's value, true carrier costs plus delay costs with the sign reversed, not its cash, and that value is what the Shapley clause of H5 compares with alone. In every tested coalition every owner's Shapley value was above its alone profit: the smallest difference was each of A and B's in coalition AB with equal shares, +54.2 M USD [29.7, 82.0], and the largest owner C's in coalition ABC with equal shares, +472 M USD [436, 508] (Table 8). Against the no-payment pooled market on the same seeds, each owner's profit changed by: under first price with equal shares A +6.5 M USD [5.0, 8.0], B +10.8 M USD [9.3, 12.3], C −14.7 M USD [−15.0, −14.5]; under first price with 50/35/15 shares A −0.9 M USD [−2.5, 0.7], B +10.2 M USD [8.7, 11.7], C −6.5 M USD [−6.6, −6.4]; under the per-item second price with equal shares A +13.3 M USD [12.5, 14.2], B +13.9 M USD [12.9, 15.0], C −14.5 M USD [−14.7, −14.3]; under the per-item second price with 50/35/15 shares A +5.9 M USD [4.9, 6.9], B +13.4 M USD [12.4, 14.5], C −6.2 M USD [−6.4, −6.1]; under the posted

price with Shapley sharing with equal shares A +119.8 M USD [103.8, 137.3], B +94.7 M USD [82.4, 107.0], C −214.4 M USD [−238.4, −190.9]; under the posted price with Shapley sharing with 50/35/15 shares A +34.3 M USD [22.6, 46.5], B +50.2 M USD [34.9, 68.6], C −84.6 M USD [−100.4, −68.7] (M USD a year).

Table 8. Each owner's Shapley value in every tested coalition under the posted price against working alone on the same seeds, in M USD a year: the alone profit, the paired difference (Shapley value in the coalition minus alone) and its 95 % bootstrap interval over seeds. In a two-owner coalition both owners receive the same difference by construction, half the coalition's gain.

Split	Coalition	Owner	n	Alone profit	Difference	95 % CI
equal shares	AB	A	30	−106.7	+54.2	[29.7, 82.0]
equal shares	AB	B	30	−118.7	+54.2	[29.7, 82.0]
equal shares	AC	A	30	−106.7	+366.2	[335.1, 399.9]
equal shares	AC	C	30	−723.4	+366.2	[335.1, 399.9]
equal shares	BC	B	30	−118.7	+353.6	[320.2, 388.0]
equal shares	BC	C	30	−723.4	+353.6	[320.2, 388.0]
equal shares	ABC	A	30	−106.7	+172.3	[145.3, 201.4]
equal shares	ABC	B	30	−118.7	+159.6	[136.0, 184.5]
equal shares	ABC	C	30	−723.4	+471.7	[436.3, 508.4]
50/35/15 shares	AB	A	30	−141.9	+65.2	[36.4, 93.9]
50/35/15 shares	AB	B	30	−119.0	+65.2	[36.4, 93.9]
50/35/15 shares	AC	A	30	−141.9	+169.4	[146.6, 193.5]
50/35/15 shares	AC	C	30	−313.0	+169.4	[146.6, 193.5]
50/35/15 shares	BC	B	30	−119.0	+163.7	[144.8, 185.2]
50/35/15 shares	BC	C	30	−313.0	+163.7	[144.8, 185.2]
50/35/15 shares	ABC	A	30	−141.9	+115.0	[89.9, 142.4]
50/35/15 shares	ABC	B	30	−119.0	+109.3	[85.6, 133.7]
50/35/15 shares	ABC	C	30	−313.0	+213.5	[191.8, 236.3]

5.7 Manipulation and audits (H5)

With equal shares the largest gain from scaling came from owner B at k = 1.10 under both bid rules: +7.73 M USD a year [−1.46, 18.53] under first price and +6.91 M USD [−2.30, 17.71] under the per-item second price; that deviation changed the fleet's cargo delay by −5,500 h [−16,272, 3,704] and the system cost by −15.07 M USD. With 50/35/15 shares the largest gain from scaling came from owner A at k = 1.25 under both bid rules: +5.23 M USD a year [−9.72, 19.31] under first price and +2.89 M USD [−12.05, 16.97] under the per-item second price; that deviation changed the fleet's cargo delay by −72 h [−11,323, 11,596] and the system cost by +2.65 M USD. Under the posted price there is no bid to scale. Under first price with equal shares an audit share of 5 % recovered 5 % of the gain, leaving +7.34 M USD and an audit share of 20 % recovered 20 % of the gain, leaving +6.18 M USD; under first price with 50/35/15 shares an audit share of 5 % recovered 15 % of the gain, leaving +4.43 M USD and an audit share of 20 % recovered 61 % of the gain, leaving +2.03 M USD; under the per-item second price with equal shares an audit share of 5 % recovered 6 % of the gain, leaving +6.53 M USD and an audit share of 20 % recovered 22 % of the gain, leaving +5.37 M USD; under the per-item second price with 50/35/15 shares an audit share of 5 % recovered 28 % of the gain, leaving +2.09 M USD and an audit share of 20 % recovered 111 % of the gain, leaving −0.31 M USD a year net of expected penalties, at a penalty multiple of 3 times the misreport. H5 clause by clause, in both splits: that first-price bids reward shading is not supported (the largest first-price gain's 95 % interval is [−1.46, 18.53] with equal shares and [−9.72, 19.31] with 50/35/15 shares, in M USD a year); that the per-item second price reduces the largest gain is supported (paired difference −0.82 M USD [−0.93, −0.70] and −2.33 M USD [−2.51, −2.13]); that the posted price with

Shapley sharing leaves no owner worse off than alone in any tested coalition is supported, against the owners-alone baseline of Section 3.5 (the comparison against no payments is in Section 5.6; over every owner and coalition the smallest upper bound of Shapley value less alone profit is +82.0 M USD and +93.9 M USD, and the smallest lower bound +29.7 M USD and +36.4 M USD). The cash part of that gain, with the allocation held fixed, was +2.57 M USD a year [2.43, 2.73] with equal shares and +5.40 M USD [5.01, 5.80] with 50/35/15 shares, a decomposition made after the runs (Section 4.6); the registered measure adds the deviator's own cargo delay, whose standard deviation over seeds is at least 7.6 times that cash.

5.8 What did not work

Every cell of E46 ran all its seeds. Every measurement of Section 4.2 was made. Every one of the 49 phase-5 checks of Section 3.6 tagged to this paper that ran passed; 9 were not run: E4 solar input (not applicable: no solar power model in the engine); X2 not possible (not possible: the browser has no ATC engine (web/sim.js is the frozen v2/v3 engine); the browser's ATC safety program is checked against the E44 fixtures by FRONTEND's tests); X5 not possible (not a computation: it needs reviewers outside the team watching replays (the owner's call)); R6 skipped (not applicable: the relay is modelled as a coverage share (B_EARTH_RELAY coverage p), not an orbit); R8 skipped (skipped by the ten-minute rule: the published values use other criteria and antenna heights, so a fair comparison needs the prior-art note's criteria matched case by case); L3 skipped (skipped by the ten-minute rule: the policy outputs a bid correction, not a cost or ETA prediction, so a calibration target would have to be defined first); V3 skipped (FRONTEND's: the scene's interpolation and the wheel height above the DEM live in the browser); V4 skipped (FRONTEND's: the drawn directions and link lines live in the browser); T4 skipped (skipped by the ten-minute rule: it needs exponential traversal and Markov routing through a new test hook in the road layer). After every experiment had run, a verification check found that cargo ids were not unique: 9–18 of about 2,970 items a year reused an earlier id, in every regime (change 10 of the pre-registration). Ids enter the simulator's decisions only through a planning memo within one dispatch call. The fix was tested by rerunning 48 full years (E40, E43: every regime or mode, seeds 1–3), of which 48 gave outcomes identical item by item, compared by content without ids; reused ids fell from 670 to 0. No experiment was rerun and no reported number changes; trace and scene files written before the fix still carry the duplicate ids. Nothing else is withheld: every cell that ran is in Table 7 and every measurement made is in Tables 5 and 6.

6. Discussion

6.1 What the results say about the question

The question had three parts: what the vehicles of 2028–2032 can carry, what Lunar ATC needs inside each of them and whether it fits, and under which payment rules a market among owners is fair and hard to game. On the first part the answer needed no experiment and is the least comfortable: the fleet's capacity is a plateau at 2,800 kg per lift, set by two cargo rovers nobody has bought, and the crew rovers NASA has bought carry 80 kg and drive 10 km on a charge. Whatever a dispatcher does, elements heavier than the plateau wait for a vehicle that does not exist, and the base's stated need runs to 15,000 kg [5, 6]. The sizing of this paper is for the fleet that will exist, which is small; the network it needs is correspondingly small.

On the second part, the safety check, the on-robot memory and the radio traffic fit the reference processor and the link class with the stated factor, the class nearest a limit being the crew rover, while one auction round in a hundred takes 1.12 s, above the one-second decision. The safety program is explicit switching by design, the form that costs least in computation in the one measured comparison [45], and the planner's learned part is 13

weights; the sizing question was always the bookkeeping of the market and the reservation table rather than any arithmetic, and the measurement says how much bookkeeping there is.

On the third part, every rule left every owner's mean profit above its alone profit; the operator ran a deficit under first price (2.50 M USD a year with equal shares), a deficit under the second price (12.71 M USD, 5.1 times first price's) and a balanced budget under the posted price; the largest gain from scaling bids was +7.73 M USD a year, and audits at 20 % did not remove it under either bid rule. A decomposition made after the runs, not pre-registered, separates the two parts of that gain: with the allocation held fixed, the scaled asks alone paid the deviator +2.57 M USD a year [2.43, 2.73] under first price with equal shares and +5.40 M USD [5.01, 5.80] with 50/35/15 shares (+1.75 M USD and +3.05 M USD under the second price); the rest of the paired gain is the change in the deviator's own cargo delay once its asks move jobs, a term with a mean of +5.16 M USD and −0.17 M USD and a standard deviation over seeds of 29 M USD and 41 M USD, at least 7.6 times the cash, which is why the registered interval covers zero although shading pays in cash. The registered test of the first clause was therefore underpowered: the measure it registered, the deviator's whole profit, carries the deviator's own delay term, whose spread over seeds is many times the cash, and the cash part on its own is positive. That is a statement about the test's power, made after the fact, not a change of the verdict, which stands as registered. The Shapley sharing of the posted price moves money from the owner of the small vehicles to the two cargo-rover owners: the Shapley transfers move 200 M USD a year from owner C with equal shares and 78 M USD a year from owner C with 50/35/15 shares to the other two owners. With equal shares the pooling gain over alone is 804 M USD a year; under no payments owner C keeps 85 % of it (A 7 %, B 8 %, C 85 %), under the Shapley sharing owner C receives the largest share, 59 % (A 21 %, B 20 %, C 59 %), and the transfer of each owner is the difference between the two (A +112 M USD, B +88 M USD, C −200 M USD); with 50/35/15 shares the pooling gain over alone is 438 M USD a year; under no payments owner C keeps 68 % of it (A 18 %, B 13 %, C 68 %), under the Shapley sharing owner C receives the largest share, 49 % (A 26 %, B 25 %, C 49 %), and the transfer of each owner is the difference between the two (A +34 M USD, B +44 M USD, C −78 M USD). Owner C's loss against no payments (Section 5.6) is the mirror image of A and B being paid for carrying C's cargo, and it follows from the fleet split, not from the payment rule: owner C holds the two small rovers and the two hoppers (Section 3.2), which cannot carry most of its own cargo, so under no payments the cargo rovers of A and B carry that cargo for nothing, under the posted price at the cost-recovery tariff, and the Shapley sharing charges C for the value those vehicles add. That is what the Shapley rule's definition of fairness means, each owner's marginal contribution averaged over the orders in which owners could join, and it is a definition, not a measurement: it gives owner C the largest share of the pooling gain over alone, and C's transfer is the difference between that share and the larger one C keeps when nobody is paid. The alone baseline is weak, because an owner alone is dominated by its cargo's delay cost at its own value of time, so every pooled rule clears it by a wide margin. The harder comparison is against the pooled market with no payments, and there against no payments owner C is worse off under every paid rule and both splits, with the interval below zero; no other owner's interval lies below zero (A's under first price with 50/35/15 shares includes it). The market's purpose in this design is not to extract the last hour of efficiency but to let three companies share eight vehicles without a treaty; the number that matters for that is whether any owner, in particular the smallest, loses by joining, and E46 measures it directly against both baselines.

6.2 Comparison with earlier studies

The radio traffic of Lunar ATC, 48.0 B/s per robot, is 287 times smaller than the 13.8 kB/s the two-rover convoy controller of [25] used; the market's own messages come to 0.34 per robot-hour at today's fleet. Nearly all of

those bytes are the status heartbeat at 1 Hz (Section 5.2 gives its share), so the honest comparison is between a one-message-a-second heartbeat and a convoy controller that exchanges state many times a second, not between a market and a convoy. The mesh itself agrees with NASA's viewsheds [16]: a 10 m mast on this ridge sees 23.3 % of the route points directly, within the 10.6–23.7 % that a 4 m tower and the 12.1–29.2 % that a 20 m tower covered of a 20 × 20 km polar area in that study (Section 2.2), and the rest of the network is multi-hop, as the Subterranean Challenge teams found underground [54, 55]. The smallest owner of the 50/35/15 split gained under the posted price with Shapley sharing (+213 M USD a year [192, 236] in the grand coalition against working alone) and lost 85 M USD against the no-payment market, where Kondor's smallest operator paid the largest fleet penalty [39]; the operator's budget under the second price was a deficit. Kondor's figure is a non-coordination penalty, the extra fleet an operator needs outside a pool, and ours is a gain over working alone inside one, so the two are not the same quantity and agree only in the direction of who fares worst. The second-price deficit is mechanical rather than a prediction of theory: the payout is floored at the carrier's own ask and the tariff is the reserve when no competitor exists, so on every job the operator pays out at least what the shipper pays in and on most jobs more.

6.3 Implications

For the companies that will own the vehicles, the implication of Section 3.2 is that the first base cannot be served by any one of them, and that the first thing worth agreeing is not a market but a coupling: two rovers of two owners lifting one load, which is the only way anything above 2,000 kg moves in this fleet [25, 60]. For the designers of those vehicles, Table 3 is the list of what a vehicle must carry to take part in a shared network: one small program that refuses unsafe moves, a radio that floods a few dozen bytes a second to whatever it can see, and a copy of the entries it has heard, and Section 5.1 says what they cost. For the agencies that write LunaNet, the surface-to-surface link that the specification leaves bilateral [15] is where every number in Section 5.3 lives; a lunar UHF band [24] would change the link budget more than any software choice. For the operator of the base, the payment rule is a policy choice with a measured price: every rule kept every owner in; the budget was a balanced budget under the posted price and a deficit under the second price. The choice is distributional: owners A and B prefer any paid rule to none and owner C prefers none, so the operator's measured price of a payment rule is owner C's loss against the no-payment market (Section 5.6). Earth's role in this design is the one it can play from 400,000 km away: it sets the prices and the rules, keeps the ledger, settles the accounts and audits a sample, and nothing it does has to happen within a radio window.

6.4 Limitations

The findings hold for one base, one fleet of eight vehicles in three ownerships, one radio model with assumed antenna heights, range and loss, and one implementation of the software; the sizing is scaled to flight hardware by a factor, not measured on it; the market has three owners, one manipulation and one private-cost spread, prices each run after one allocation with no owner responding to a rule, and tests individual rationality against a weak alone baseline; the Shapley transfers are a definition of fairness, not a measurement; and H5's decision rules were written after the registration, before any run. Section 7 states each in full.

6.5 Future work

Four pieces follow directly. The safety program and the auction should be run on the flight-class processor itself, against recorded traffic from the companion experiments, to replace the scaling factor with a measurement. The link budget should be checked against a measured lunar surface link as soon as one exists; none does today [17].

The market should be run with five owners and with the manipulations this paper dropped, false availability, charger hogging and cherry-picking, which an auditable ledger may or may not catch. And the Phase-1 rovers' real battery, mass and radio should replace the assumed values when their design reviews publish them, due in late 2026 [1].

7. Limitations

The sizing is scaled, not flown. Every processor time and memory figure is measured on a laptop in Python and scaled by the ratio of a reference flight processor's published benchmark to the laptop's. The factor is stated with a range, but a flight implementation in another language on another operating system could differ from it by more than that range in either direction. The factor used is 54.8 (range 36.5–64.0), from Intel Core i5-10500H (6 cores, 2.5 GHz base, 4.5 GHz turbo) to the Frontgrade Gaisler GR740 (rad-hard quad-core LEON4FT, 250 MHz). The laptop's benchmark is itself assumed: no published CoreMark exists for its processor, and the figure is 6.0 CoreMark/MHz times its 4.5 GHz turbo clock (Assumed, range 4–7 CoreMark/MHz), so the factor's range is the range of that assumption. The processor share reported for H10 takes the 99th-percentile call time at every call, which overstates the mean load.

The auction round and the safety loop are not allocated to cores. The reference processor has four cores, and the paper does not say which of them runs the 10 Hz safety loop and which the auction round. On a shared core, a round that exceeds one second, which Section 5.1 reports at the 99th percentile, would stall the safety check while it ran; the design must give the safety check its own core or bound the round's time, and this paper states that requirement and leaves it open.

The radio is a model. Line of sight on a 5 m terrain grid with assumed antenna heights, a fixed range and a fixed loss probability stands in for a radio that nobody has measured on the Moon; the link budget uses standard diffraction and two-ray models over a regolith whose permittivity is unmeasured [20, 21]. The share of route sample points in direct view of the mast is a property of this model on this ridge, and it is a share of points, not of time.

The fleet is the fleet of 2028–2032 as published in October 2026. The cargo rovers are uncontracted, their masses, batteries and one capacity are assumed, and the Phase-1 rovers' batteries are set so that the published 10 km per charge comes out [1], because no battery size is published. The design reviews due in late 2026 and early 2027 may change every assumed value in Table 2.

Three owners, one manipulation. The market is tested with three owners in two splits; the theory's warnings about collusion and shill identities [31] concern exactly such thin markets and are not tested. Bid scaling is the only misreport simulated; an owner that misreports position, availability or job completion, or that hogs a charger, is not. The audits compare bids with logged costs and assume the logs are honest.

H5's decision rules are dated after the registration. The first-price cell that H5's first two clauses need was dropped on 27 October and restored on 29 October, before any E46 row (change 7 of [59]); the numeric decision rules for all three clauses were written then, not in the original registration, and are dated in [59].

Private costs are assumed and their spread is not swept. Each owner's cost of energy, wear and time is the engine's job cost times a lognormal multiplier with $\sigma = 0.2$, a distribution we chose; profits and individual rationality are measured against those costs. The declared range of 0.1–0.3 was not run, because each value needs a new year of physical runs, so the results hold at one spread, and no spread replaces a company's own accounting.

Payments are priced after one allocation. Each rule prices a run that the auction has already allocated, so no owner responds to a rule and the honest allocation is the same under all four rules. The manipulation test is one deviator at a time over four fixed factors; no best response, no joint deviation and no equilibrium is computed. The gains reported are what a fixed scaling would have earned, not what a strategic owner would find.

The efficiency loss differs between the splits. The market's efficiency loss against the oracle differs between equal and 50/35/15 shares (Table 7); nothing in this paper traces that difference to a cause, and it is left untraced.

The alone baseline is weak. An owner alone is dominated by its cargo's delay cost at its own value of time, so individual rationality against alone is easy to pass under every rule; the comparison against the pooled market with no payments is the harder one, and Section 5.6 reports it.

The Shapley transfers are a definition, not a measurement. The transfers under the posted price are large relative to the cash the tariff moves, and they follow from the Shapley rule's definition of fairness; the paper measures what that definition pays each owner, not whether the owners would call it fair.

The first-price clause failed for power. The registered measure, the deviator's whole profit, carries the deviator's own cargo delay, whose spread over seeds is many times the cash the scaled asks earn; the cash part is positive (Section 6.1, post hoc). The clause is not supported as registered, and that is not evidence that shading does not pay.

The first-price budget is a calibration gap. With honest asks the first-price budget is the tariff receipts less the true carrier cost; the tariff was set to the mean cost per tonne-kilometre on the calibration seeds, so the budget on the test seeds is the gap between the two seed sets, not a property of the rule.

The lander carries only the radio unit and the landing announcement. In this design a lander's own Earth link is for flight; its Lunar ATC radio unit broadcasts the landing announcement on the mesh, Earth also holds the time, and the lander is otherwise a closure; it carries no planner and no safety program (change 7 of [59], Table 3), and the radio unit and announcement part it does carry are not sized here, because E48 measures the on-robot part on the driving and flying classes.

8. Conclusion

This paper asked what the vehicles of 2028–2032 can carry, what a shared traffic and market network must put inside each, whether it fits, and under which payment rules distrustful owners gain by joining. We built the vehicle table from published sources, described the radio network, measured our implementation at one to eight times the fleet, and ran 1,080 simulated years under four payment rules with bid scaling and audits. The fleet's capacity is a plateau at 2,800 kg per lift, set by two uncontracted cargo rovers. On flight hardware the safety check and radio traffic fit the reference processor and link for 20 robots; one auction round in a hundred takes 1.12 s, above the one-second decision. Every owner beat alone under every rule, but owner C lost against no payments under every paid rule, and the largest gain from scaling bids was +7.7 M USD a year, interval including zero. A post hoc split shows that gain's cash part positive and its delay part noisy.

For practice, Table 3 lists what a vehicle needs to join a shared network, at a measured cost; for policy, the surface link that LunaNet leaves bilateral is where the design's numbers live, and the payment rule is a choice priced in budget and in who pays whom. The results hold for one base, one small fleet, a modelled radio and a scaled, not flown, implementation. The vehicles are known, the network can be small, and the market can be priced.

Data and code availability

The simulation engine, the vehicle catalogue with every source (`lunarsim/atc/vehicles-p5.json`), the mesh data product and its summary (`data/atc/connecting_ridge/mesh-v1.json.gz`, `results/ATC_mesh.json`), the safety program (`lunarsim/atc/safety.py`), the auction and message layers, the regime defaults (`lunarsim/atc/regimes.py`), the experiment runner, every results file this paper quotes (`results/E46_*`, `results/E48_sizing.json`) and the script that builds the paper from them (`paper/p3/build_p3.py`) are in the repository at github.com/azizjonrahimov/Lunar-Haul. The pre-registration with its dated changes is `paper/atc/PREREG.md` [59]. The live site at azizjonrahimov.github.io/Lunar-Haul shows the vehicles with a "Lunar ATC inside" layer, the base, the network and a walk-through of the market.

Author contributions

A.R. set the questions, the scope of the three papers and the standards of evidence, directed the design of the vehicle table, the network model, the sizing measurements and the market experiment, reviewed the hypotheses, the results and the text, and is responsible for the content.

Use of AI tools

Large-language-model assistants (Anthropic's Claude, used through Claude Code) wrote the simulation code, the safety program, the auction and message layers, the instrumentation, the experiment runner and the analysis scripts, ran the experiments, drafted and revised the text and compiled the reference list, all under the author's direction and review. Every number in the paper is written by the build script from the results files or the catalogue and summary files named in the text, and each sentence that states a direction or a verdict is checked by that script against the data; no number was typed from an assistant's output. The references were collected from published sources and checked against the sources listed in the repository's research notes. The author takes responsibility for the whole.

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No external funding was received. The author founded Roatify, which proposes a traffic and dispatch network for lunar vehicles of the kind sized here. The pre-registration, the reported assumptions and the build-script checks are the safeguards offered against that interest.

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